Aarush Prasad Uli

Seattle, WA | 425-499-6472 | aarushprasad@hotmail.com
Socials: github.com/aarushprasad47 | linkedin.com/in/aarush-uli-143839271 | devpost.com/aarushprasad
Website: https://personal-website-9uzs71qc9-aarushprasad47s-projects.vercel.app

Education

University of Washington, Seattle — Expected June 2028
B.S. Computer Science, GPA: 3.78 / 4.0 — *Seattle, WA*

- Relevant Coursework: Data Structures, Algorithms, Systems Programming, Databases, Hardware/Software Interface, Machine Learning, Deep Learning
- Selected for the Lavin Entrepreneurship Program (Autumn 2025 cohort), a competitive program within the Buerk Center for Entrepreneurship and Foster School of Business

Experience

Durran Lab — Jan. 2026 – Present
ML & Data Pipelines Research — UW Atmospheric Sciences & Geophysics — *Seattle, WA*

- Contributed to the Deep Learning Earth System Model (DLESyM), a GPU-accelerated Earth system simulation developed in collaboration with *NVIDIA*; advised by Prof. Dale Durran and PhD mentor Nathaniel Cresswell-Clay
- Engineered **Python** data pipelines processing **multi-TB** model output on HPC clusters, computing verification diagnostics (atmospheric blocking statistics, seasonal-cycle metrics, ENSO indices)
- Built reusable analysis and visualization tooling in **NumPy** and **PyTorch** to quantify model skill against reanalysis datasets; adopted by faculty and research staff for ongoing model validation

Four Elements Engineering — Sep. 2023 – Aug. 2025
Software Engineering Intern — *Redmond, WA*

- Automated field-measurement ingestion and recurring analysis workflows as scheduled **Apache Airflow** DAGs, with pandas-based ETL loading cleaned data into **PostgreSQL**, cutting manual processing time by ∼**80%**
- Implemented data-quality validation with **Pydantic** schemas and **pytest** test suites, catching malformed or out-of-range sensor readings before they reached production models
- Built physics-based hydrological models in **Python** (**NumPy**, **SciPy**, **pandas**) generating predictive stormwater runoff data for **3+** civil infrastructure projects across the Puget Sound region

Pack of Parts Robotics — Sep. 2023 – June 2025
Software Engineer — Embedded Software & Controls — *Redmond, WA*

- Developed swerve-drive firmware in **Java** implementing closed-loop motor control with real-time encoder feedback under hard timing constraints; tuned PID loops for autonomous navigation
- Integrated an **OpenCV** vision pipeline for real-time target acquisition, fusing vision estimates with encoder state to drive an actuated arm, improving scoring efficiency by **55%**

Projects

OmniAge — Real-Time Triage Platform | *Spring Boot, Django, Kafka, Swift/SwiftUI, SQLite, Gemini API* — Sep. 2026 – Present

- Architected an event-driven backend for a native iOS triage platform: a **Django** REST service ingests and normalizes patient data from multiple sources and publishes record events to **Apache Kafka** topics, handling **100+** simulated records
- Built the triage engine as a **Spring Boot** microservice consuming Kafka events, maintaining a dynamic priority queue that re-ranks records by acuity and schedules follow-ups from **Gemini**-parsed free-text notes, reducing triage time by a projected ∼**40%**
- Designed RESTful APIs and the **SQLite** persistence layer; Kafka-driven queue updates keep a **SwiftUI** operator dashboard synchronized in real time, surfacing high-priority alerts as new data arrives

ShelfLife — IoT Food Freshness Monitor | *ESP32, SHT31, JavaScript, MongoDB, 3D Printing* — Oct. 2026

- Built an end-to-end IoT prototype at *DubHacks '26*: an **ESP32** with an **SHT31** temperature/humidity sensor streams storage-condition readings to a **MongoDB** backend, housed in a custom 3D-printed enclosure
- Developed an Machine Learning risk engine that estimates food spoilage risk from cumulative exposure history and per-food profiles using a **Perceptron** model, rather than relying on point-in-time temperature checks
- Built a **JavaScript** dashboard unifying sensor history, inventory, and freshness-risk insights, with estimates designed to stay useful despite gaps in sensor data

Technical Skills

Languages: Java, Python, C++, C, Swift, C#, Rust, JavaScript/TypeScript, SQL
Backend & Data: Spring Boot, Azure, .NET, Django, Apache Kafka, Apache Airflow, REST APIs, PostgreSQL, SQLite
Cloud & Tools: AWS (S3, Bedrock, DynamoDB), Git, Linux, React, Flutter, SwiftUI
ML & Data Science: PyTorch, NumPy, pandas, SciPy, OpenCV, Gemini API
Spoken Languages: English (native), Japanese (fluent), Spanish (fluent), Hindi